Lecture : Linear Codes (Contd.) and Examples Thereof

In this lecture, we shall continue with our study of structured, linear codes that are subspaces of $\mathbb{F}^n$, for some finite field $\mathbb{F} = \mathbb{F}_q$ and blocklength n.

We first revisit two simple codes we had seen much earlier:

(i) Repetition code : Let $\mathbb{F} = \mathbb{F}_2$. The repetition code $\mathcal{C}^{REP}$ consists of just two codewords: $\{\underline{0}, \underline{1}\} \subseteq \mathbb{F}_2^n$.

Note that $\mathcal{C}^{REP}$ is an $[n, 1, n]_2$ linear code over $\mathbb{F}_2$.

(ii) Single parity-check code : Let $\mathbb{F} = \mathbb{F}_2$. The single parity-check code $\mathcal{C}^{SPC}$ is the collection:

$$\mathcal{C}^{SPC} = \{\underline{c} : \sum_{i=1}^{n} c_i = 0\}.$$

Claim 1: $\mathcal{C}^{SPC}$ is a linear code with $\dim(\mathcal{C}^{SPC}) = n-1$.

Proof: Linearity is easy to check: indeed, pick $\underline{c}_1, \underline{c}_2 \in \mathcal{C}^{SPC}$ and note that $\underline{c}_1 + \underline{c}_2 \in \mathcal{C}^{SPC}$. By a fact about subspaces we had seen earlier, it follows that $\mathcal{C}^{SPC}$ is linear.

Note also that there are exactly 2^{n-1} codewords in $\mathcal{C}^{SPC}$, since there are as many sequences of "even parity", i.e., the # of 1s in

which is even. Hence, it follows that $\dim(\mathcal{C}^{SPC}) = n-1$. $\blacksquare$

<u>HW</u>: What is $d_{min}(\mathcal{C}^{SPC})$?

<u>Def 1</u>: A "generator matrix" G for an $[n,k]_q$ code $\mathcal{C}$ is a $k \times n$ matrix over $\mathbb{F}_q$ whose rows form a basis for $\mathcal{C}$, i.e.,

$$G = \begin{bmatrix} \leftarrow & g_1 & \rightarrow \\ \leftarrow & g_2 & \rightarrow \\ & \vdots & \\ \leftarrow & g_k & \rightarrow \end{bmatrix}$$

where $\mathcal{C} = \text{span}(g_1, g_2, \dots, g_k)$.

<u>Remarks</u>: (i) Note that $\text{rank}_{\mathbb{F}}(G) = \dim_{\mathbb{F}}(\mathcal{C})$.

(ii) Since there are likely to be several bases for $\mathcal{C}$, there will be several generator matrices for $\mathcal{C}$.

We use generator matrices for encoding a message to a codeword in the following way:

An encoder for $\mathcal{C}$ is a 1-1 map from message vectors $\underline{m} \in \mathbb{F}_q^k$ to codewords $\underline{c} \in \mathcal{C}$. Since every codeword $\underline{c} \in \mathcal{C}$ can be uniquely expressed as $\underline{c} = \sum_{i=1}^{k} \alpha_i \cdot g_i$, given G, we define the encoder to be the mapping

$$\underline{m} = (m_1, \dots, m_k) \xmapsto{\mathcal{E}_G} \underline{m}\, G = \sum_{i=1}^{k} m_i g_i .$$

Remark: Note that since G is full rank, the mapping above is 1-1. Specifically, $\mathcal{E}_G(\underline{0}) = \underline{0}$, for all generator matrices G.

Def 2: A generator matrix G for a linear code $\mathcal{C}$ is "systematic," if it is of the form

$$G = [\underline{I}_k \mid B],$$

where $\underline{I}_k$ is the $k \times k$ identity matrix.

Remark: (i) Via encoding using a systematic generator matrix, we obtain that

$$\underline{m} \xmapsto{\mathcal{E}_G} [\underline{m} \mid \underline{m}B]$$

first k symbols of the codeword constitute the message itself. These are called "information symbols."

(ii) Not every code has a systematic generator matrix. Indeed, consider the code

$$\mathcal{C} = \left\{ \begin{matrix} 0000 \\ 0001 \\ 0100 \\ 0101 \end{matrix} \right\}$$

Note that $\mathcal{C}$ is a $[4, 2, 1]$ linear code. The first symbol is 0 in all codewords; hence $\nexists$ a systematic generator matrix for $\mathcal{C}$.

Def 3: A "parity-check matrix" H for an $[n,k]_q$ linear code $\mathcal{C}$ is an $(n-k)\times n$ matrix over $\mathbb{F}_q$ such that $\mathcal{C} = \text{nullspace}(H)$, i.e.,

$$\mathcal{C} = \{\underline{c} : H\underline{c}^T = \underline{0}\}.$$

Theorem 1: Given a generator matrix G and a parity-check matrix H for a linear code $\mathcal{C}$ over $\mathbb{F}_q$, we have that

$$H \cdot G^T = [\underbrace{\underline{0}}_{\text{all-zeros matrix}}],$$

i.e., $H g_i^T = \underline{0}$, $\forall i \in [k]$.

Proof: Since $\mathcal{C} = \text{nullspace}(H)$, and since $g_i \in \mathcal{C}$, $\forall i \in [k]$, we have that $H g_i^T = \underline{0}$, $\forall i \in [k]$. $\blacksquare$

FACT: Via the rank-nullity thm. and the linear-algebraic fact that a basis of a finite-dimensional vector space is a spanning set of the smallest size, we obtain that the rows of H are linearly independent, i.e., that H is full rank.

HW: Suppose that $\mathbb{F} = \mathbb{F}_2$ and that $G = [I_k | B]$ for a linear code $\mathcal{C}$. Obtain a parity-check matrix for $\mathcal{C}$.

Example: The $[7,4,3]_2$ binary Hamming code. Consider the code $\mathcal{C}$ with parity check matrix

$$H = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

← coll^n of all non-zero vectors in $\mathbb{F}_2^3$ as columns, but permuted to make the code systematic. →

HW: Verify that $\dim(\mathcal{C}) = 4$ and that $d_{min}(\mathcal{C}) = 3$.

Theorem 2: Let H be a parity-check matrix for an $[n,k]_q$ linear code $\mathcal{C}$. Then, $d_{min}(\mathcal{C})$ is the smallest integer $d>0$ such that <u>every</u> collection of $d-1$ columns of H is linearly independent over $\mathbb{F}_q$.

Proof: We wish to prove the statement that:

"$d_{min}(\mathcal{C})$ is the smallest integer $d>0$ such that <u>some</u> collection of d columns of H is linearly independent over $\mathbb{F}_q$."

If some coll^n of d cols. of H is linearly dependent, then $\exists$ codeword $\underline{c} \in \mathcal{C}$ of wt. $\leq d$. $\Rightarrow d_{min}(\mathcal{C}) \geq d$.

Further, if d is the smallest such integer, there exist <u>no</u> codewords in $\mathcal{C}$ of wt. $< d$. $\Rightarrow d_{min}(\mathcal{C}) \leq d$. ∎

HW: Via Theorem 2, identify what it means for a linear code to have minimum distance 1 (or 2). Likewise, derive what it means for a linear code to be 1-error correcting, i.e., to have distance 3 or more.

Dual Codes

Associated with any given linear code $\mathcal{C}$ is the dual code $\mathcal{C}^{\perp}$, defined as follows:

Def 4: The dual code $\mathcal{C}^{\perp}$ of a linear code $\mathcal{C}$ of blocklength n over $\mathbb{F}_q$ is

$$\mathcal{C}^{\perp} \triangleq \{\underline{x} \in \mathbb{F}_q^n : \underline{x} \cdot \underline{c} = 0, \text{ for } \underline{\text{all}} \ \underline{c} \in \mathcal{C}\}.$$

"dot product", i.e., $\underline{x}_1 \cdot \underline{x}_2 = \sum_{i=1}^{n} x_{1,i} \cdot x_{2,i}$.

over $\mathbb{F}_q$

The following theorem then holds.

Theorem 3: Let G be any generator matrix of $\mathcal{C}$. Then,
$\mathcal{C}^{\perp} =$ nullspace(G), i.e., G is a parity-check matrix for $\mathcal{C}^{\perp}$.

Proof: Exercise

Via the rank-nullity theorem, we obtain that $\dim(\mathcal{C}^{\perp}) = n - \dim(\mathcal{C})$.

Examples: ① If $\mathcal{C}$ is the repetition code $\mathcal{C}^{REP}$ with parity-check matrix $G = [1 \ 1 \ 1 \ldots 1]$, then its dual code is

$$\mathcal{C}^{REP,\perp} = \left\{ \underline{x} \in \mathbb{F}_2^n : \sum_{i=1}^{n} x_i = 0 \pmod 2 \right\}$$
$$= \mathcal{C}^{SPC}.$$

<u>Theorem 4</u>: We have that $(\mathcal{C}^{\perp})^{\perp} = \mathcal{C}$.

<u>Proof</u>: (i) $\mathcal{C} \subseteq (\mathcal{C}^{\perp})^{\perp}$: To see this, note that for any $\underline{c} \in \mathcal{C}$, we have $\underline{c} \cdot \underline{x} = 0$, for all $\underline{x} \in \mathcal{C}^{\perp}$.

Now, since $\dim((\mathcal{C}^{\perp})^{\perp}) = n - \dim(\mathcal{C}^{\perp}) = \dim(\mathcal{C})$, we must have that $\mathcal{C} = (\mathcal{C}^{\perp})^{\perp}$. $\blacksquare$

The above arguments also help show that any parity-check matrix H of $\mathcal{C}$ is a generator matrix of $\mathcal{C}^{\perp}$.

<u>HW</u>: Recall the definition of the $[7,4,3]_2$ Hamming code. We now define the family $\{\mathcal{C}_r^{HAM}\}_{r \geq 2}$ of Hamming codes, where

$\mathcal{C}_r^{HAM}$ is the length-(2^r-1) code with parity-check matrix

$$H_r^{HAM} = \begin{array}{c} \\ r \\ \\ \end{array} \overbrace{\begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ \underline{h}_1 & \underline{h}_2 & \cdots & \underline{h}_{2^r-1} \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}}^{2^r-1},$$

where $\{\underline{h}_1, \dots, \underline{h}_{2^n-1}\}$ are the (distinct) non-zero vectors in $\mathbb{F}_2^n$.

What is the dimension and minimum distance of $\mathcal{C}_n^{HAM}$? How many errors can it correct?